

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2023-2024

Comprehensive Examination
(EC-3 Regular)

Course No. : DSECLZG522
Course Title : Big Data Systems
Nature of Exam : Open Book
Weightage : 40%
Duration : 2.5 Hours
Date of Exam : EN

No. of Pages	= 2
No. of Questions	= 7

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made, if any, should be stated clearly at the beginning of your answer.

Q.1. In each of the following scenarios, point out and give a brief reason what type of multi processor computer one would use as per Flynn's taxonomy, i.e. the choices are SIMD, SISD, MIMD, or MISD.

(a) A scientific computing application does a $f_1(x) + f_2(x)$ transformation for every data item x given f_1 and f_2 are specialised operations built into the hardware.

(b) A video is processed to extract each frame which can be either an anchor frame (full image) or a compressed frame (difference image w.r.t anchor). A compressed frame (C) is transformed using a function f , where each pixel is compared with the last anchor (A) to recreate the uncompressed image (B), i.e. $B(i, j) = f(C(i, j), A(i, j))$ for all pixels (i, j) in the input frames.

(c) A multi-machine Apache Hadoop cluster system for data analysis.

(d) A development system with multiple containers running JVMs and MongoDB nodes running on a single multi-core laptop.

(5 Marks)

Q.2. Examine the Spark scala code given below:

```
val lines = new Array[String](3)
lines(0) = "zero one reject"
lines(1) = "one two reject"
lines(2) = "two two zero"
val stringRDD=sc.parallelize(lines,2)
println("Number of Partitions: "+stringRDD.getNumPartitions)
val wordsRDD = stringRDD.flatMap(x => x.split(" "))
val filteredRDD = wordsRDD.filter(x => x.startsWith("o") || x.startsWith("t"))
val wcRDD = filteredRDD.map(word => (word, 1)).reduceByKey(_ + _)
wcRDD.collect().foreach(println)
```

1. What is the output of this code ?
2. Describe the RDD lineage in the above code

(5 Marks)

Q.3. Describe the most appropriate scheduling mechanism to be configured with Hadoop YARN in the following scenarios:

(a). A company wants to run jobs of equal size on a Hadoop cluster. The number of jobs can vary at any point in time.

(b). A company wants to run short jobs and long jobs on the same cluster and short jobs cannot always be starved by long jobs.

(c) A company wants to divide the Hadoop cluster resources in different proportions between the departments and does not care about wastage of unutilized resources.

(d) A company wants to run jobs of different nature concurrently on the same cluster and is worried about wastage of resources.

(5 Marks)

Q.4. You are given access to 10 Terabyte of data with details of daily sales transactions belonging to an e-commerce company. The transactional data files are stored in a folder on HDFS with a structure as shown below:

Transaction No. (Unique)	Invoice	Stock Code	Description	Quantity	Invoice Date	Price(\$)	Customer ID	Country
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It is not required to modify the data. You are not allowed to copy or move the data into any other storage location. Select a suitable analytics tool from the Hadoop ecosystem family to run queries on this data. Give justifications for your selection. Using the tool selected by you, write a query to find the total sales revenue received in \$ from each country.

(7 Marks)

Q.5. Why is windowing needed in Stream computing? Data from a smart device is transmitted over TCP/IP connection to a Spark streaming analytics system. Data from the device arrives at the rate of 1 packet per second. The batching interval is set to 5 seconds. The stream analytics system uses a stateful algorithm with a Time window of 60 seconds. It is required to generate an output from the streaming system every 30 seconds. What type of windowing is recommended by you? Give appropriate configuration parameters for the Window suggested by you. What will be the number of RDDs in the Dstreams generated in one window duration.

(7 Marks)

Q.6. An eCommerce company is planning to launch a new application on cloud for online sale of items. During normal seasons of the year, this application deployed on a cluster of 4 servers will guarantee acceptable response to the customers. During festival seasons, the sales is expected to double. Which feature of cloud will be useful in minimizing the data center costs without compromising the response time. What are the recommended features of the component products to be used in the design of this application?

(6 Marks)

Q.7. Explain in detail about storage considerations in Big Data

(5 Marks)
